DESCRIPTIVE ANALYTICS REPORT

AND

IBM COGNOS PRACTICALS

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Descriptive Analytics Report: Financial Performance and User Engagement Trends

Tracking Revenue, User Growth, Regional Performance, Monetization, User Behaviour and Trend Patterns

Note on data:-

Pinterest is a visual discovery and social media platform that enables users to discover, save, and organize ideas through images, videos, and links known as "*Pins*." Since its launch in 2010, Pinterest has become one of the **world's leading digital platforms for inspiration in areas such as fashion, home décor, food, travel, education, and lifestyle**. The platform has become an **important digital marketing channel for businesses, content creators, and brands** because it helps them reach audience who are actively searching for inspiration and products.

As a result, Pinterest generates a large amount of user engagement data that can be analysed to understand content performance and user behaviour. Over the past five years, Pinterest has experienced significant developments in terms of revenue growth, user engagement, monetization strategies, and regional expansion. As digital advertising and e-commerce continue to evolve, understanding Pinterest's business performance provides valuable insights into how social media companies create value while maintaining user satisfaction.

This study presents a descriptive analysis of Pinterest's performance over the five-year period from 2021 to 2025. The analysis focuses on **key performance indicators, including financial performance, user growth, monetization, regional performance, growth trends, and user behaviour**. Descriptive statistical techniques and graphical representations are used to summarize and interpret the **available data, providing a clear understanding of Pinterest's overall business performance and provide insights that support better content planning and marketing decisions**

Note:- The dataset used in this study is a simulated dataset created for academic purposes to demonstrate descriptive statistical analysis.

Objectives

- How has Pinterest's **Financial performance** has changed over the past five years?
- Which **geographical region** is generating the most revenue for the platform?
- Pinterest's profitability growth compare with its revenue growth over the study period?
- What trends can be seen in **Average Revenue Per User** and state the **monetization** rate?
- Which quarter has got the rise in the **Active Users** the most over past five years? Observe the pattern seen in the **User growth**?
- What is the overall **User Engagement and it's growth** over past five years?
- How effectively are Pinterest's content impressions being converted into user engagements**?**
- State the relationship between **engagement** and **impression** (Content Reach) numbers?
- How do **User interactions** (saves, pin clicks, and outbound clicks) differ across content categories?
- How does financial performance differ from user growth over the study period?

How This Would Work At Real Scale

Processing “millions of data” is an engineering problem before it’s an analytics one. *A realistic architecture:-*

Layer	Purpose	Typical tools
Investigation	Explore, clean, and understand the dataset	Python, Pandas, Jupyter/Google Colab
Processing/Aggregation	Clean, transform, and summarize data into useful metrics	Python, Pandas, SQL
Storage	Store datasets and processed data	Excel, CSV, Google Drive
Analysis	Calculate KPIs, identify trends, and compare categories	Excel, Python, Pandas
BI/Visualization	Present findings through charts, graphs, and dashboards	Excel, Power BI

The report below reflects the output of that pipeline --- daily aggregated tables --- not raw event-level data, since no dashboard renders millions of individual rows usefully.

1.] Financial Performance & Trend Pattern

Over the simulated past five year record (2021-2025), it is observed that Pinterest's financial growth has a positive surge generating **~ $2580 M in 2021 to ~$4520 M in 2025** total revenue, with **75.19%** growth rate indicating a **strong upward trend** in the company's overall business performance.

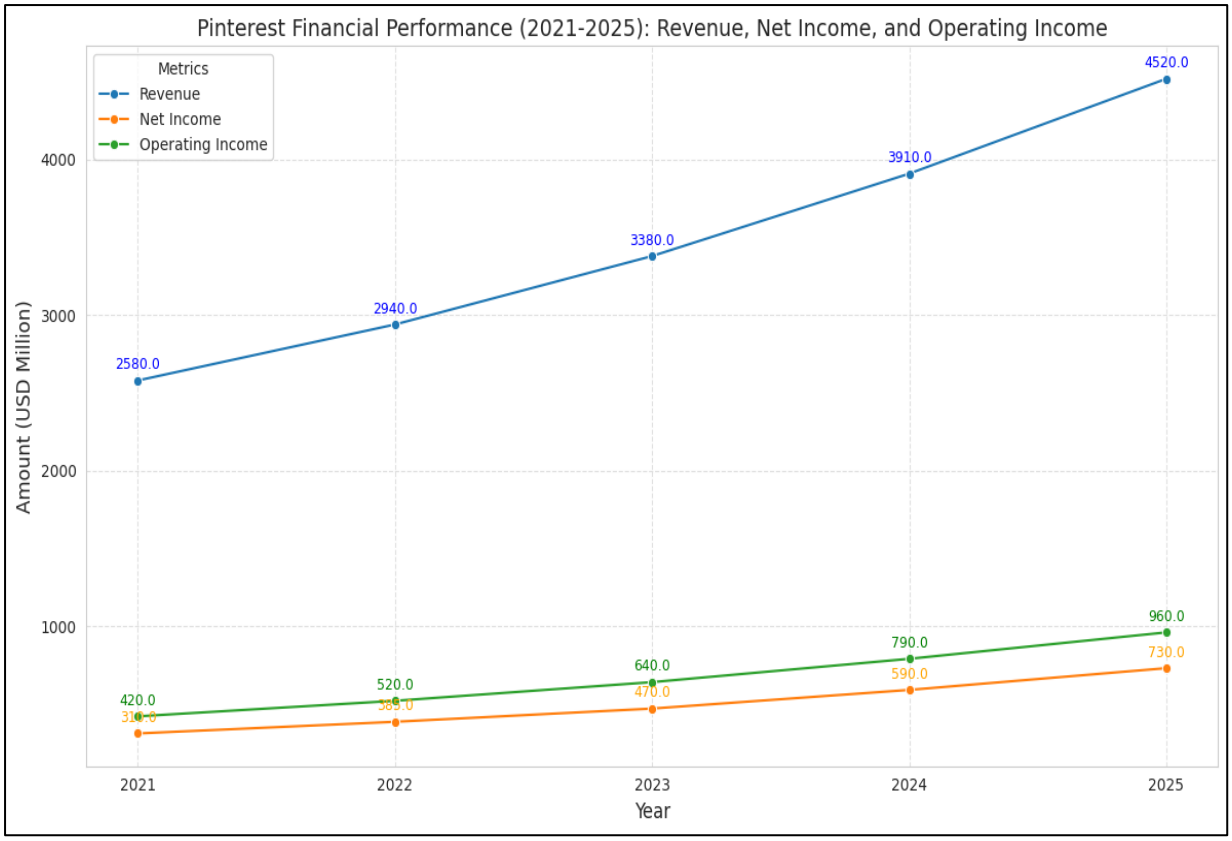
Where as, net income has also made a significant growth, rising from ~ **$310 M in 2021 to ~ $730M in 2025**, representing an overall growth of approximately **135.48%** in past five year. This substantial increase indicates a strong improvement in the company's profitability and financial performance.

Similarly, operating income increased from **~ $420 M in 2021 to ~960 M in 2025**, reflecting improved operational efficiency and stronger earnings from the company's core business activities.

KPIs Used:

- **Total Revenue=** Total annual income earned across past year period
- **Net Income=** Income left after all expenses
- **Operating Income=** Income from core business operations

Growth rate% = (Final value – Initial value/ Initial value)* 100

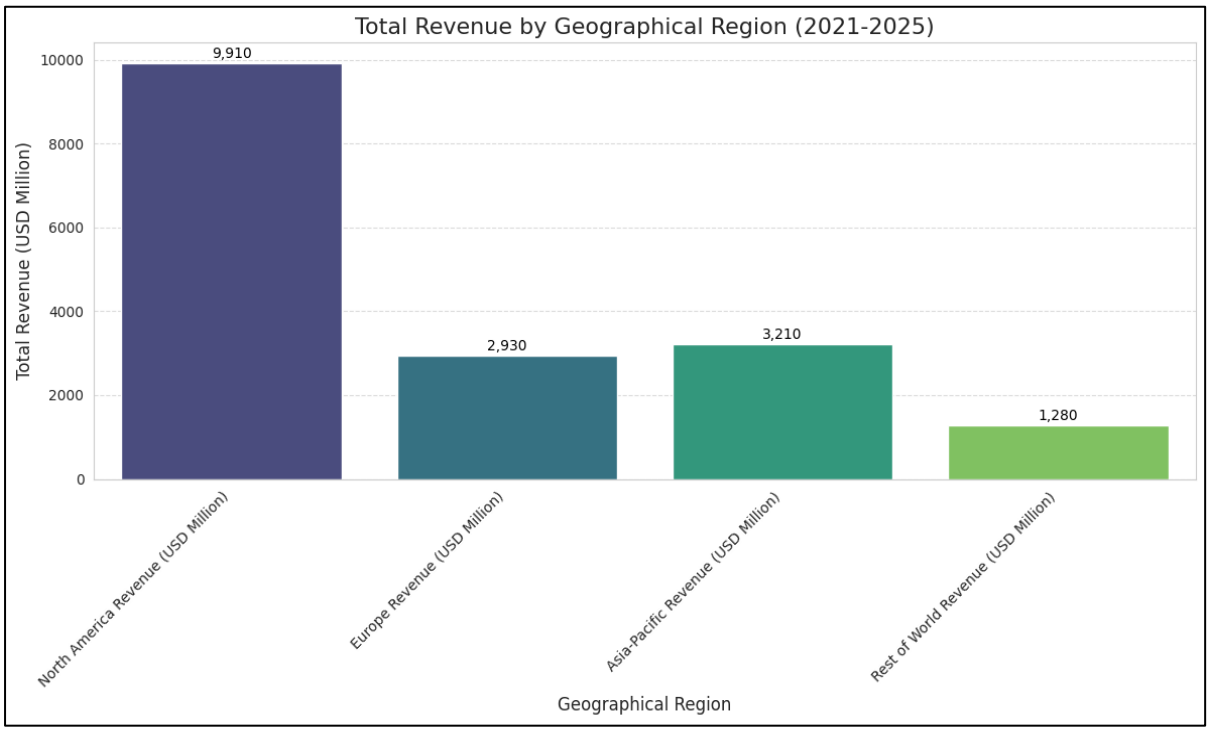

Pinterest Financial Performance (2021-2025): Revenue, Net Income, and Operating Income
Metrics
Revenue
Net Income
Operating Income
Amount (USD Million)
Year
2021
2022
2023
2024
2025
1000
2000
3000
4000
2580.0
2940.0
3380.0
3910.0
4520.0
420.0
520.0
640.0
790.0
960.0
470.0
590.0
730.0

2.] Geographical Regions and Performance

It is observed that **North America** was the platform's **highest revenue generating region** with **~ $9910 M** according the past five year record, becoming a significant strength for the company's business performance and market.

Where as, the Asia-Pacific region ranked second in revenue generation with **~ $3210 M** followed by Europe ranking third generating a revenue of **~ $2930 M**, while in comparison the Rest of the World contributed the lowest revenue of **~ $1280 M**.

Despite the differences in revenue levels, all regions demonstrated steady year-over-year growth, indicating an overall expansion of Pinterest's global business operations.



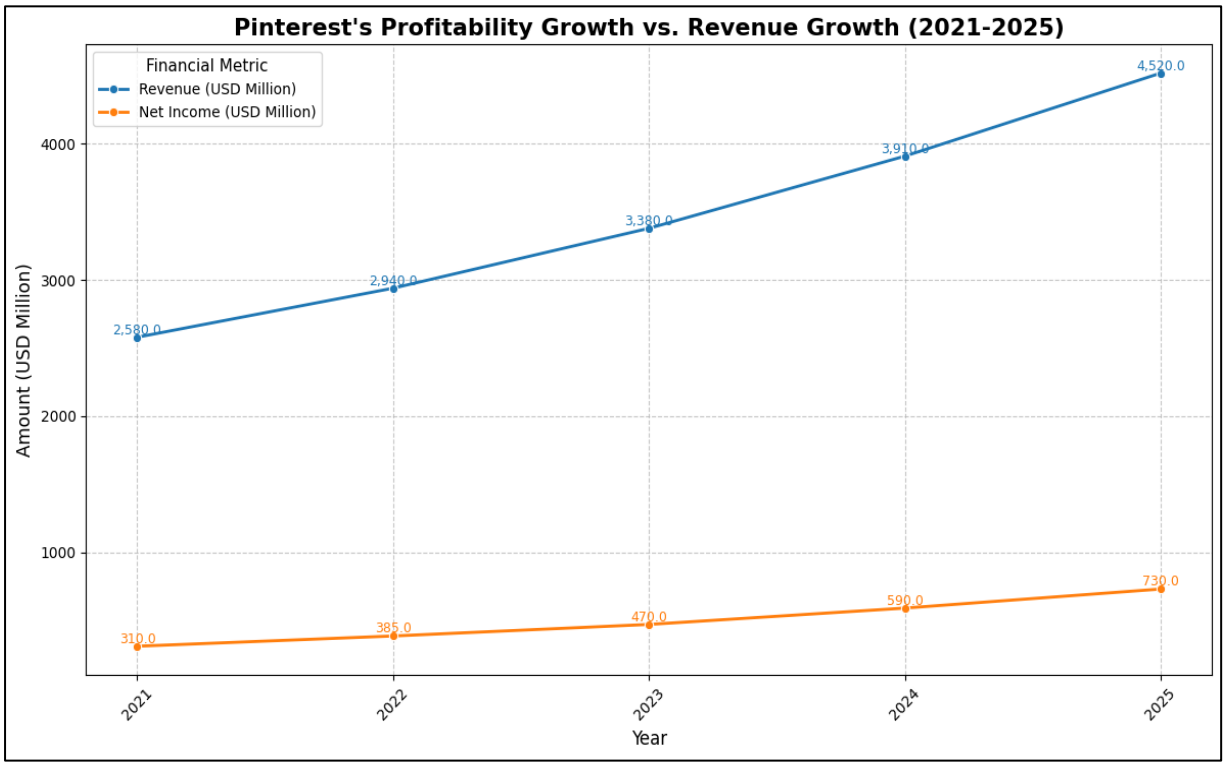
3.] Profitability growth v/s Revenue growth

It is observed that profitability grew at a substantially faster rate than its revenue during 2021–2025. Revenue increased from **$2,580 million in 2021 to $4,520 million in 2025**, representing **75.19%** growth. In comparison, Operating Income increased from **$420 million to $960 million (128.57%)**, while Net Income increased from **$310 million to $730 million (135.48%)**.

This indicates that although revenue grew steadily, Pinterest's profitability improved at a considerably higher rate over the study period.

KPIs Used:-

- **Revenue Growth**
- **Net Income Growth**



4.] Annual Analysis of Monthly Active Users and Monetization

A steady increase is seen in the monthly active users from **445 million** active users recorded in 2021 reached to **545 million** active users in 2025, having a growth rate of **22.47%**, reflecting a thriving growth of an upward trend in platform's visibility and reach.

While the Average Revenue Per User also has an upswing, generating ~ **$5.80 in 2021 to ~ $8.29** in 2025 with overall growth rate of **42.93%** , giving a clear view of **positive trend with increased expansion**.

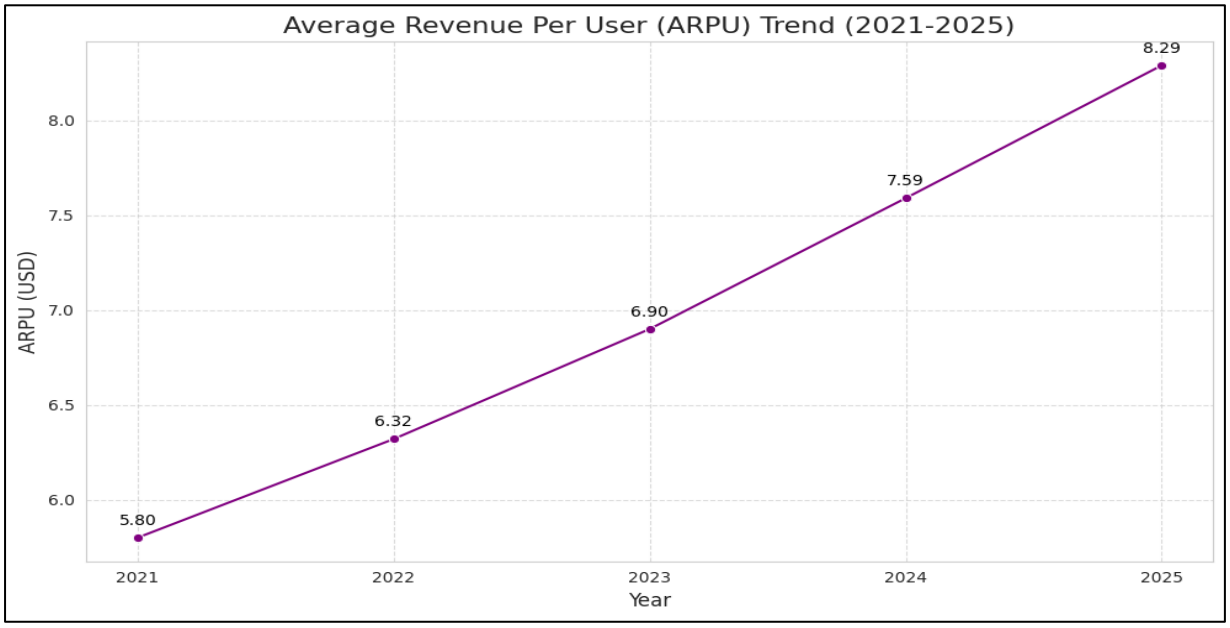
KPIs Used:-

- **Monthly Active Users=** The number of unique users who are active on the platform at least once during a month.

  Annual MAUs= (Q1+Q2+Q3+Q4)/4

- **Average Revenue Per User (ARPU) =** Measures one aspect of monetization (the overall concept of converting user traffic into revenue)

  ARPU= Revenue (USD Million) / Monthly Active Users (Million)

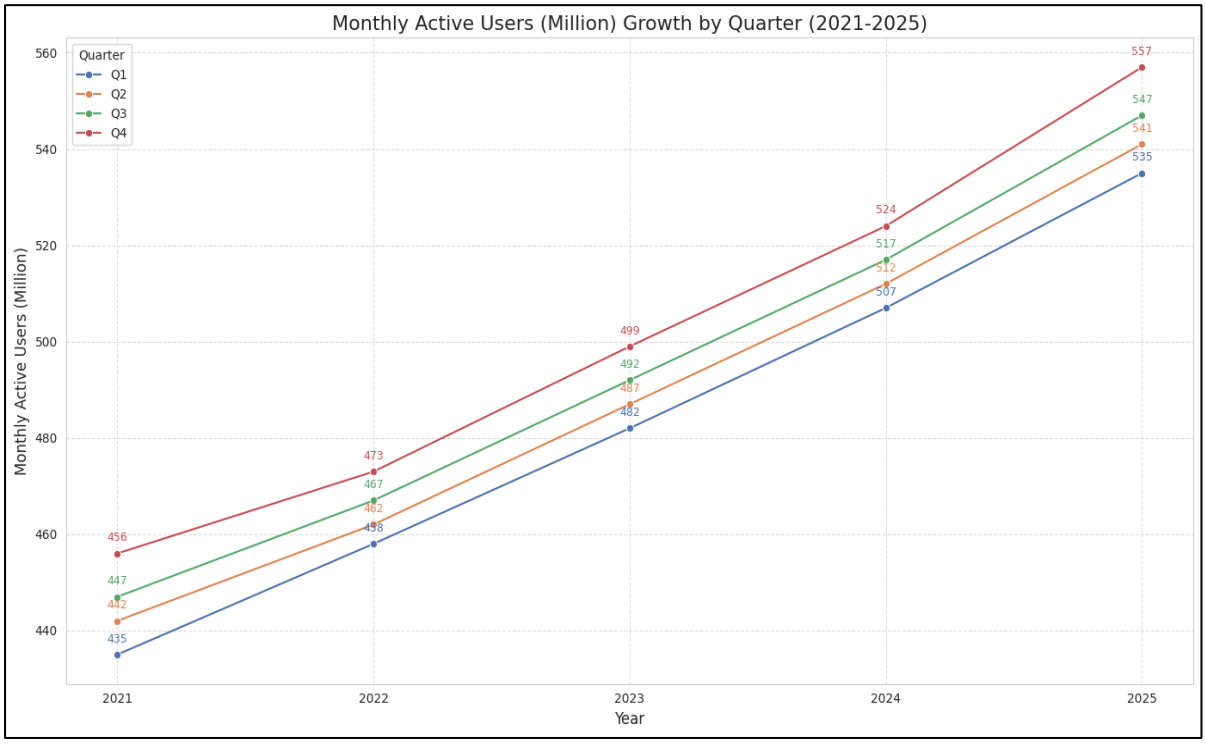


5.] Quarterly Analysis of MAU and User Activity

Overall, the findings suggest that the user activity was highest during **Quarter 4** in each year of the study period with an growth rate of **22.15%**, indicating a recurring seasonal pattern in the platform's user growth. This pattern may be associated with increased seasonal planning and higher digital marketing during the holiday period.

KPIs Used:-

- **Monthly Active Users**
- **Overall Growth rate%**



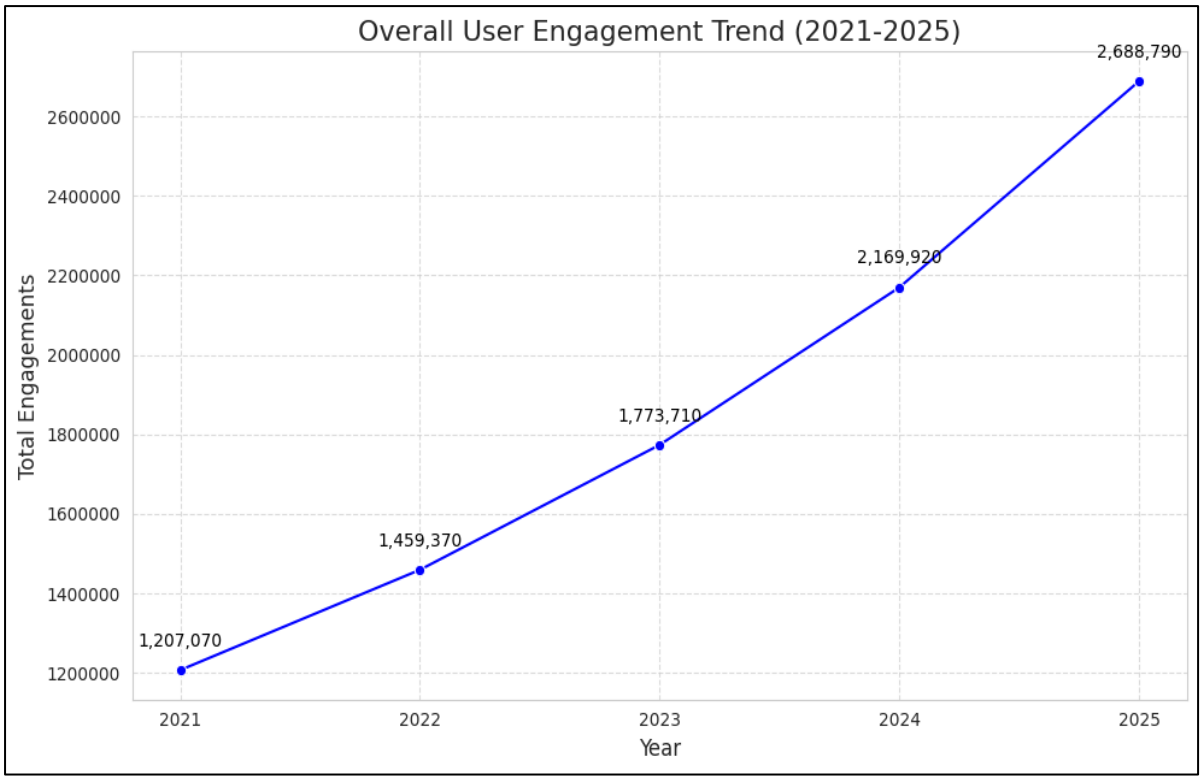
6.] User Engagements

It is observed that the platform has a boost in the overall user engagements, reflecting a strong upward trend over the past five year record. Total engagements increased from **1,207,070 in 2021 to 2,688,790 in 2025** with an overall growth of **122.75%**. Year-over-year growth remained consistently **above 20%**, indicating a steady increase in the user interaction with the Pinterest content.

KPIs Used:-

- **Total User Engagement=** The total number of user interactions with Pinterest content, including actions such as saves, pin clicks, and outbound clicks.

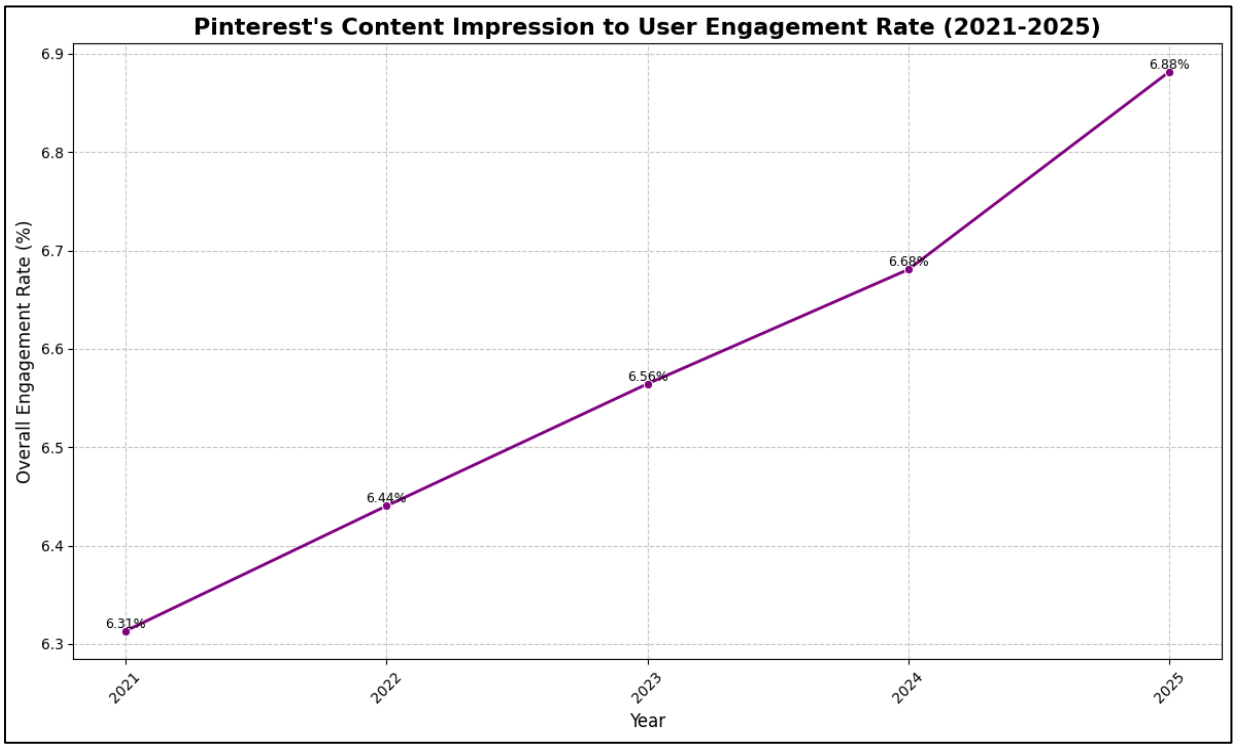
Growth= (Final engagement-Initial engagement / Initial engagement)* 100



7.] Content Impressions To User Engagements

It is seen that the platform's content visibility and user engagement both increased from **2021 to 2025**. Total impressions increased from **19.12 million to 39.07 million**, representing **104.34% growth**, while total engagements increased from **1.21 million to 2.69 million**, representing **122.75% growth**. Although impressions remained substantially higher than engagements, engagements grew at a faster rate.

Consequently, the overall engagement rate increased from **6.31% in 2021 to 6.88% in 2025**, indicating a **gradual improvement in the effectiveness of converting content visibility into user engagement**.



8.] Relationship between engagement and impression

It is seen that both impression and engagement exhibited a positive upward trend throughout the study period. While impressions increased by **104.34%**, engagements grew at a slightly higher rate of **122.75%**, resulting in a modest improvement in a engagement rate from **6.31% in 2021 to 6.88% in 2025**, reflecting an overall engagement rate of **9.03%**, an increase in the proportion of impressions that resulted in user interactions.

This suggest that content visibility expanded over time and user interactions increased at a even faster pace, indicating improved audience engagement.

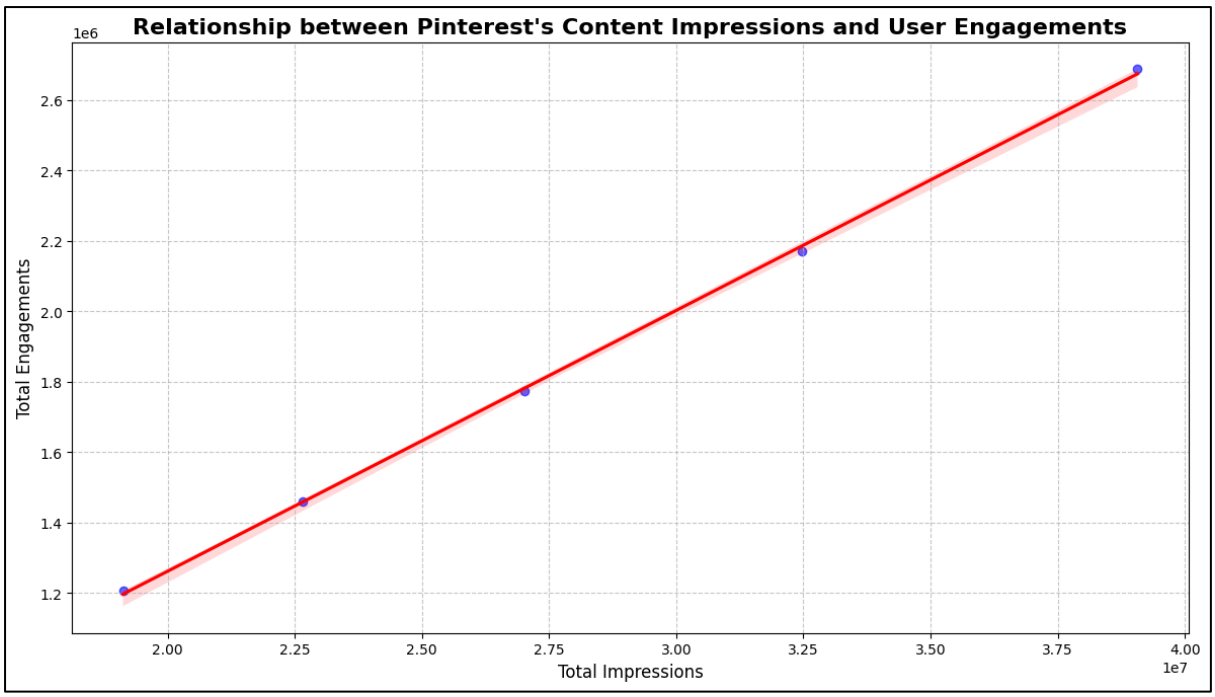
KPIs Used:-

- **Overall Engagement Rate%=** The percentage of impressions that resulted in user interactions, indicating how effectively the content engaged the audience.

Engagement Rate (%)= (Total Engagements / Total Impressions)* 100

- **Total Impressions=** The total number of times Pinterest content (Pins) was displayed to users across all selected topics during a specific period, regardless of whether users interacted with it.

Growth= (Final impression-Initial impression/ Initial impression)* 100

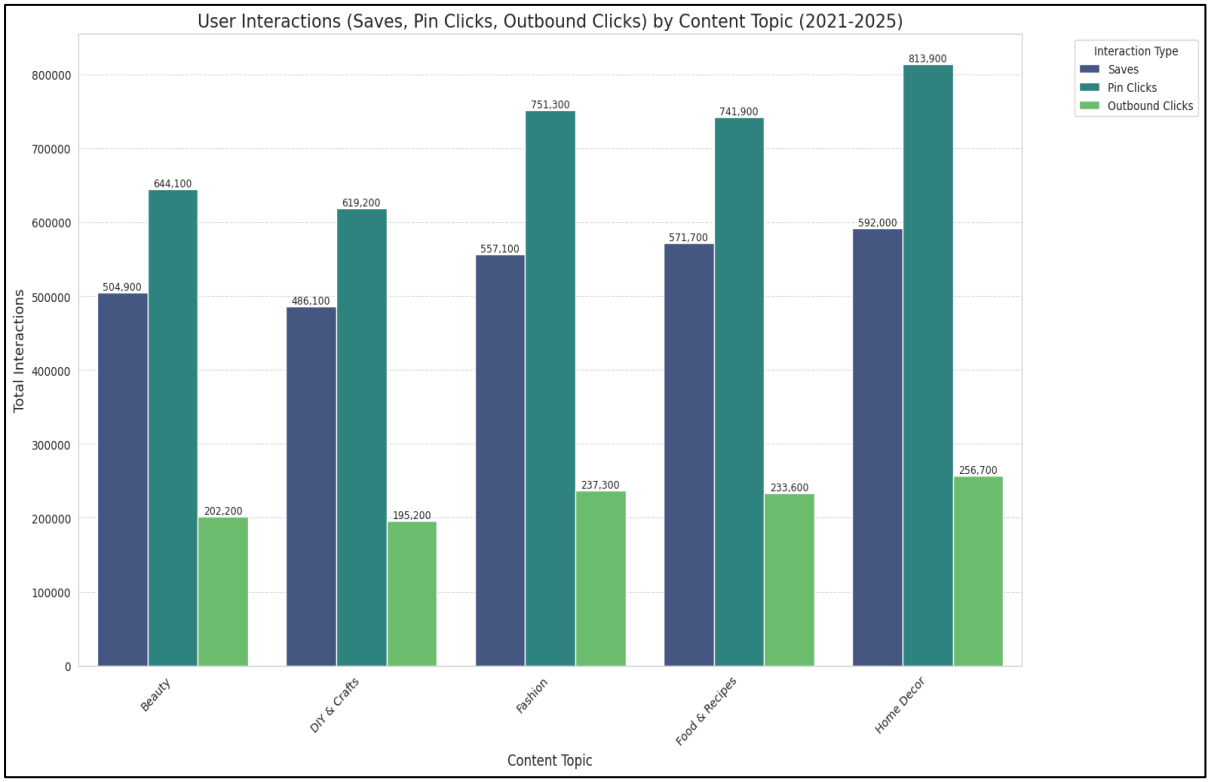


9.] User Interaction across different content

It is observed that **Home Décor** was the highest-performing content category on the platform during the 2021-2025 study period. It recorded the highest cumulative **Saves (592,000), Pin Clicks (813,900), and Outbound Clicks (256,700)**, demonstrating strong user interest, interaction, and referral traffic.

In comparison, Fashion and Food & Recipes followed with consistently strong interaction levels, although their performance remained below that of Home Decor. In contrast, Beauty and DIY & Crafts generated comparatively fewer user interactions across all three metrics.

Overall, the results suggest that user preferences and interaction behaviour differ across content categories where users were more likely to save, explore, and visit external links from Home Decor content, generating the greatest audience response.



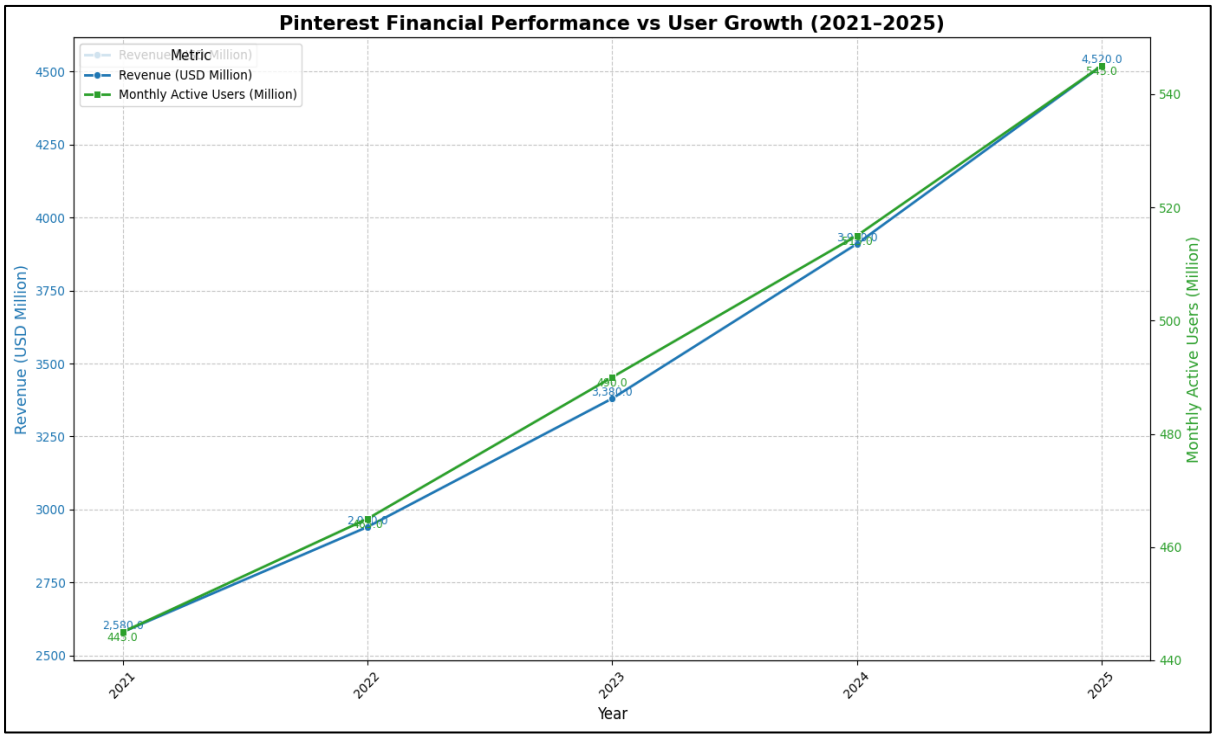
10.] Comparison between Financial performance and User growth

It is observed that Pinterest's financial performance grew considerably faster than its user base during **2021–2025**. Revenue increased from **$2,580 million in 2021 to $4,520 million in 2025**, representing a **75.19% growth**, while Net Income increased from **$310 million to $730 million**, showing a **135.48% growth**.

In comparison, Monthly Active Users increased from **445 million to 545 million**, representing a **22.47% growth**. Overall, the findings indicate that while Pinterest experienced steady user growth, its financial performance improved at a substantially higher rate over the study period.

KPIs Used:-

- **Revenue Growth Rate (%)**
- **MAU Growth Rate (%)**



Summary of Key Metrics

Growth value of metrics during the study period of past five years (2021-2025)

Metric	Value
Revenue Growth (%)	75.19%
Net Income Growth (%)	135.48%
Monthly Active Users Growth (%)	22.47%
Total Impressions Growth (%)	104.34%
Total Engagements Growth (%)	122.75%

Next Steps to Move From Descriptive ⟶ Predictive

This report answers what happened. The natural next layer, if useful:

- Based on the trends observed from **2021–2025**, Pinterest's financial performance, active user base, content impressions, and user engagement are expected to continue showing an **upward trend** in the near future.
- Content categories such as **Home Decor**, which consistently showed strong user interactions, may continue to perform well.
- The overall upward trend may continue, although the **rate of growth may vary**. These predictions are based on historical trends and should not be considered definitive future outcomes, as actual results may be influenced by **emerging trends, user interests, and changes in digital marketing practices**.

APPENDIX

1.Overview of the Datasets

The study uses multiple datasets prepared in Microsoft Excel to examine Pinterest's financial performance, user growth, monetization, content reach, and user engagement over the selected study period. The datasets were organized according to the requirements of the individual research objectives and subsequently analyzed using descriptive analysis and graphical visualization in Google Colab.

Dataset	Study period	Main Variables	Unit of Measurement	Purpose of Analysis
Financial Performance Dataset	2021-2025	Revenue, Net Income, Operating Income, ARPU, Monthly Active Users, Regional Revenue	USD Million, USD Million Users	To examine financial performance, profitability, monetization, and user growth
Quarterly Active Users Dataset	2021-2025	Year, Quarter, Active Users	Million Users	To identify quarterly patterns and determine which quarter experienced the greatest

				rise in active users
Content Performance Dataset	2021-2025	Year, Topic, Impressions, Engagements, Engagement Rate, Saves, Pin Clicks, Outbound Clicks	Number of Impression/ Interactions, Percentage	To examine content reach, user engagement, engagement efficiency, and user interactions across content categories

2.Data Analysis Tools

The datasets were organized using **Microsoft Excel**. The data was subsequently imported into **Google Colab**, where Python-based analysis and visualization were used to generate graphs relevant to the selected objectives.

The analysis primarily involved:

- Descriptive comparison
- Trend analysis
- Growth analysis
- Engagement-rate analysis
- Relationship analysis
- Category-wise comparison
- Graphical visualization

3. Note on the Dataset

The datasets used in this study were structured specifically for the purpose of descriptive analysis. The data was organized into annual, quarterly, and content-category formats according to the requirements of the selected objectives. The units and variables were kept consistent within each dataset to facilitate meaningful comparison and visualization.

Link of the Dataset used for the analysis :- Pinterest_Descriptive Analysis Report_Dataset.xlsx

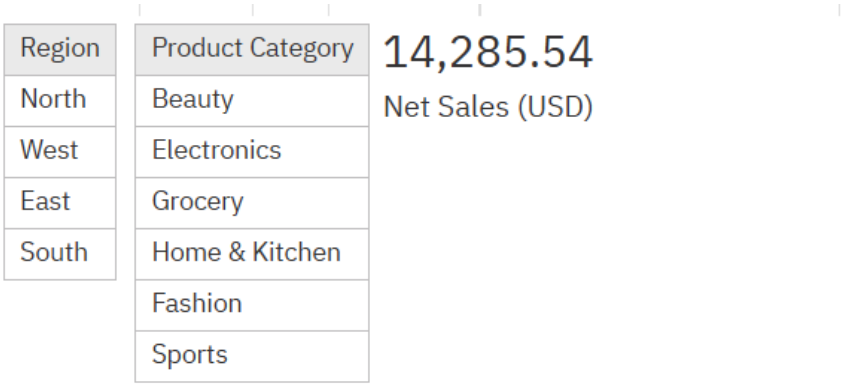
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IBM COGNOS REPORT & DASHBOARD

PRACTICALS

1) Practical 1: Generate a First Retail Pulse Report

Objective: Navigate the Cognos Analytics Reporting environment and generate a running list report against the Retail Pulse Sales Package.



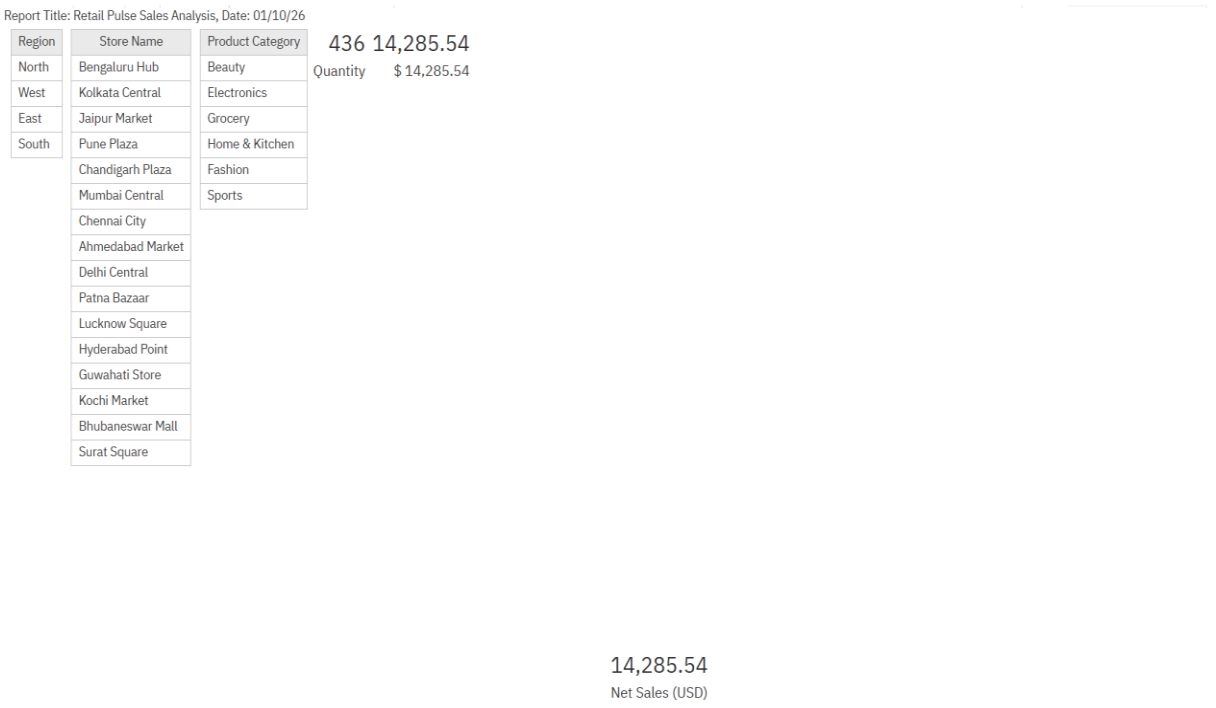
Region	Product Category	14,285.54
North	Beauty	Net Sales (USD)
West	Electronics	
East	Grocery	
South	Home & Kitchen	
	Fashion	
	Sports	

Result:- The report canvas displays live Retail Pulse rows rather than placeholder text; columns are correctly labeled; the report is saved and reopens with the same live data connection.

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2) Practical 2: Build a Grouped Retail Pulse Sales List Report

Objective: Construct a list report, group it meaningfully, format it's columns, and adds a summarizing footer.



Report Title: Retail Pulse Sales Analysis, Date: 01/10/26

Region	Store Name	Product Category
North	Bengaluru Hub	Beauty
West	Kolkata Central	Electronics
East	Jaipur Market	Grocery
South	Pune Plaza	Home & Kitchen
	Chandigarh Plaza	Fashion
	Mumbai Central	Sports
	Chennai City	
	Ahmedabad Market	
	Delhi Central	
	Patna Bazaar	
	Lucknow Square	
	Hyderabad Point	
	Guwahati Store	
	Kochi Market	
	Bhubaneswar Mall	
	Surat Square	

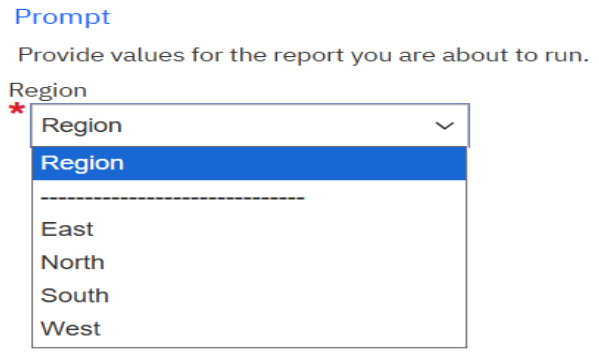
436 14,285.54
Quantity $ 14,285.54

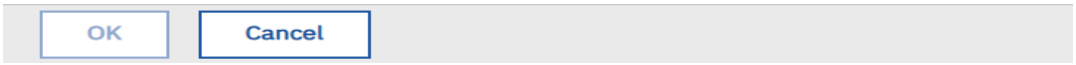
14,285.54
Net Sales (USD)

Region:- Rows are grouped by Region and Store Name with duplicates collapsed correctly; Sales Amount is formatted and right- aligned; group totals sum correctly; the page header appears on page.

3) Practical 3 : Filter Retail Pulse Sales by Region and Date

Objective: Apply a simple filter and an advanced detail filter to scope a report to a single region and reporting period, then generalize it with a prompt.





Region	Store Name	Product Category	14,285.54 Net Sales (USD)	Order Date
South	Bengaluru Hub	Beauty		27-Jul-26
	Kolkata Central	Electronics		20-Aug-26
	Jaipur Market	Grocery		13-Sep-26
	Pune Plaza	Home & Kitchen		
	Chandigarh Plaza	Fashion		
	Mumbai Central	Sports		
	Chennai City			
	Ahmedabad Market			
	Delhi Central			
	Patna Bazaar			
	Lucknow Square			
	Hyderabad Point			
	Guwahati Store			
	Kochi Market			
	Bhubaneswar Mall			
	Surat Square			

Result:- Only rows matching both filter conditions remain; the row count is visibly smaller than the unfiltered report; the prompted version correctly returns a different region's data when re-run with a new value.

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4) Practical 4 : Build a Retail Pulse Region x Product Crosstab

Objective: Construct a crosstab report summarizing Retail Pulse Sales across Region and Product Category with multiple measures and summary totals.

Net Sales (USD)	Electronics	Fashion	Grocery	Beauty	Sports	Home & Kitchen
Region	North	East	North	East	North	North

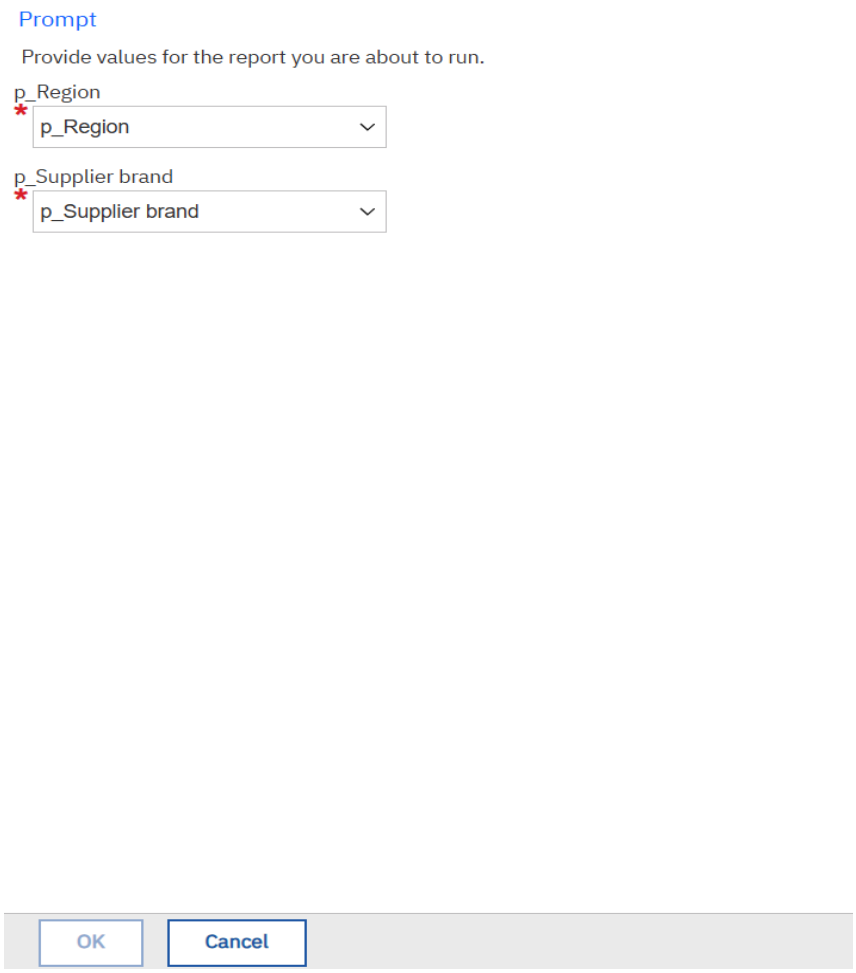
436
Quantity

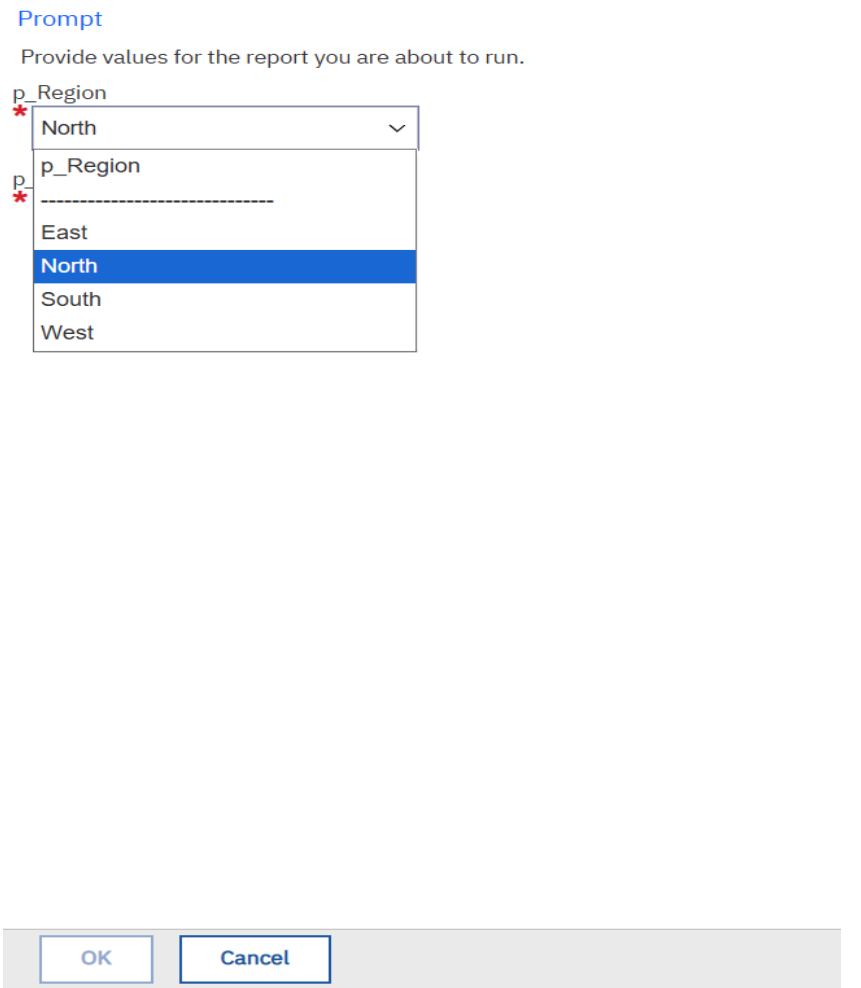
Result:- Regions and product categories align correctly without misalignment; each cell shows the correct aggregated measures; row and column totals sum correctly; Sales Amount is consistently formatted as currency.

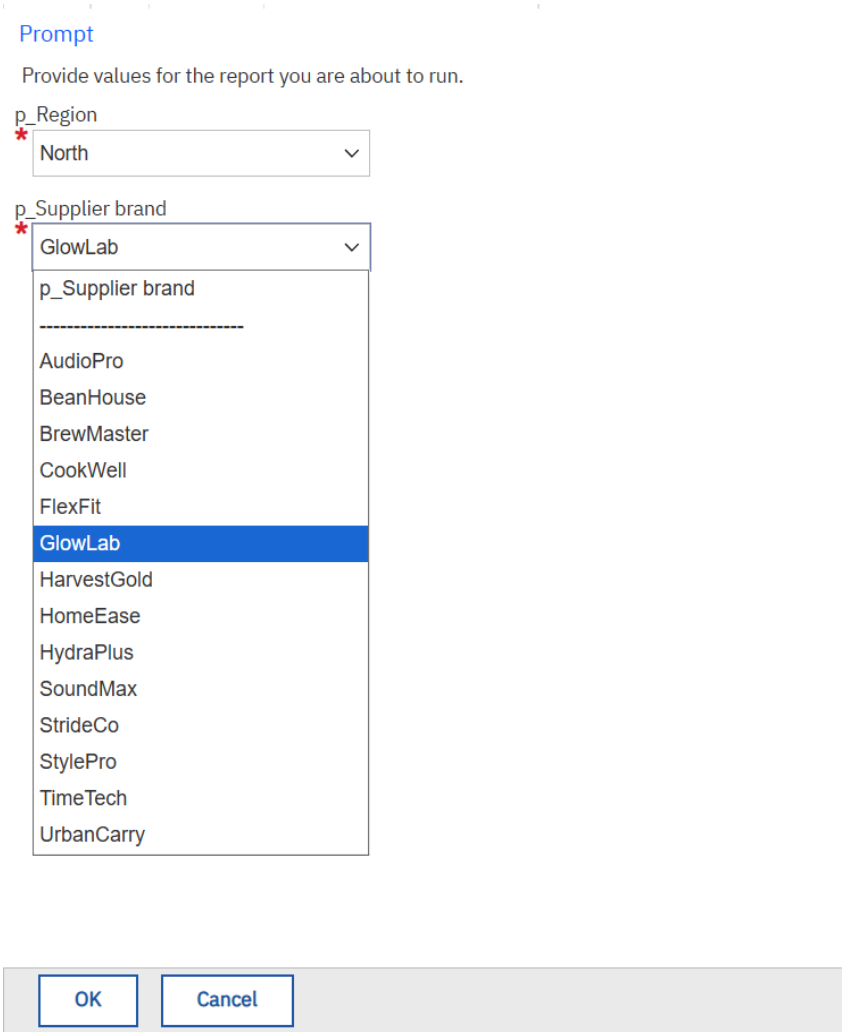
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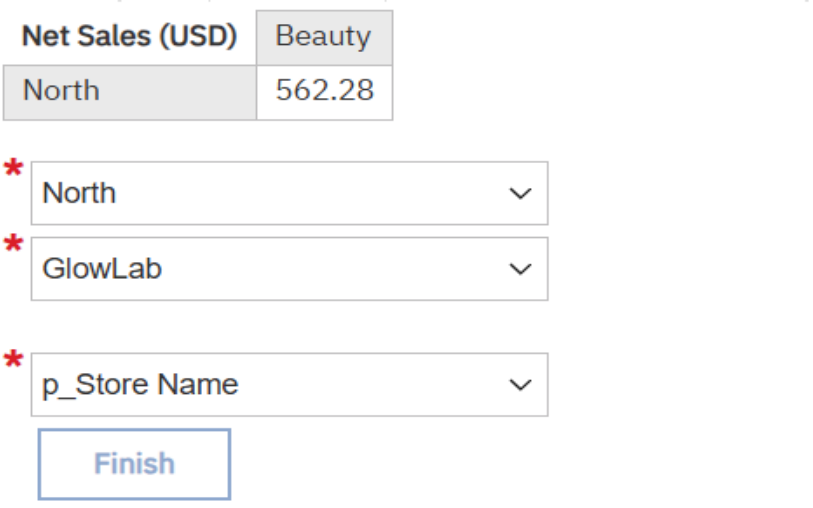
5) Practical 5 : Build a Retail Pulse Prompt Report

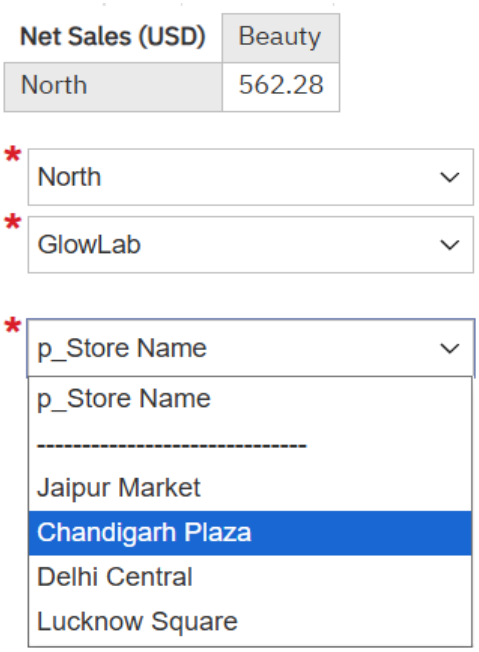
Objective: Design a prompt page with value, and cascading prompts, and link each control to a report filter.


Prompt
Provide values for the report you are about to run.
p_Region
p_Region
p_Supplier brand
p_Supplier brand
OK
Cancel


Prompt
Provide values for the report you are about to run.
p_Region
North
p_Region
East
North
South
West
OK
Cancel


Prompt
Provide values for the report you are about to run.
p_Region
North
p_Supplier brand
GlowLab
p_Supplier brand
AudioPro
BeanHouse
BrewMaster
CookWell
FlexFit
GlowLab
HarvestGold
HomeEase
HydraPlus
SoundMax
StrideCo
StylePro
TimeTech
UrbanCarry
OK
Cancel


Net Sales (USD)
Beauty
North
562.28
North
GlowLab
p_Store Name
Finish

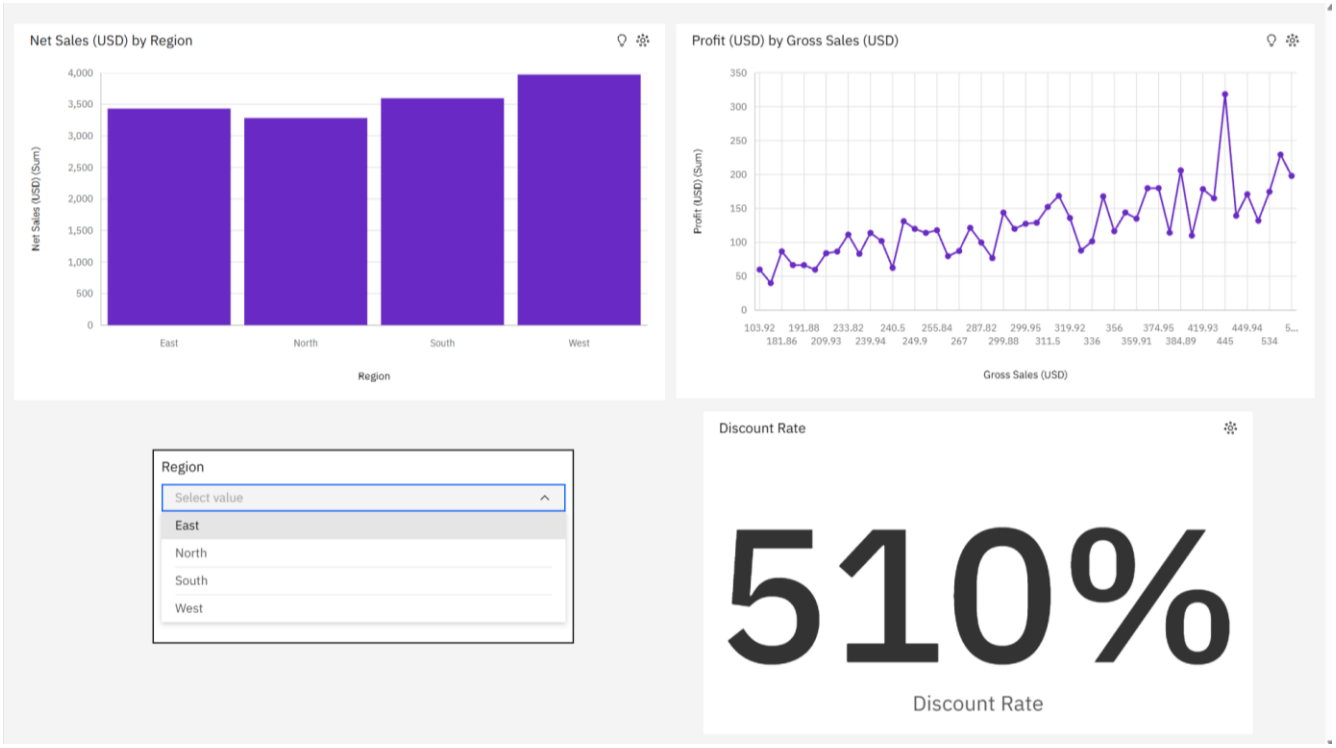


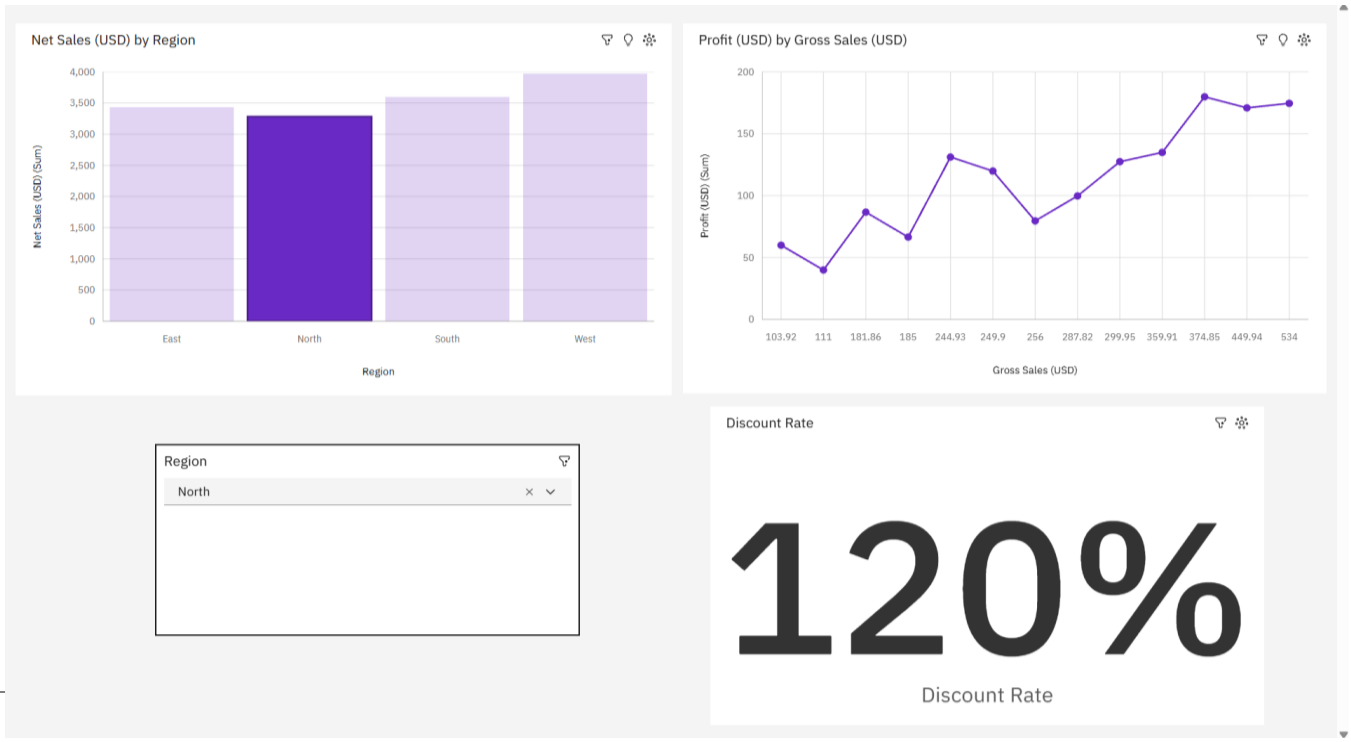
Result:- The prompt page renders all three controls correctly; changing Region updates the Store list without error; submitting returns a correctly filtered crosstab; default values produce a sensible report without any changes**.**

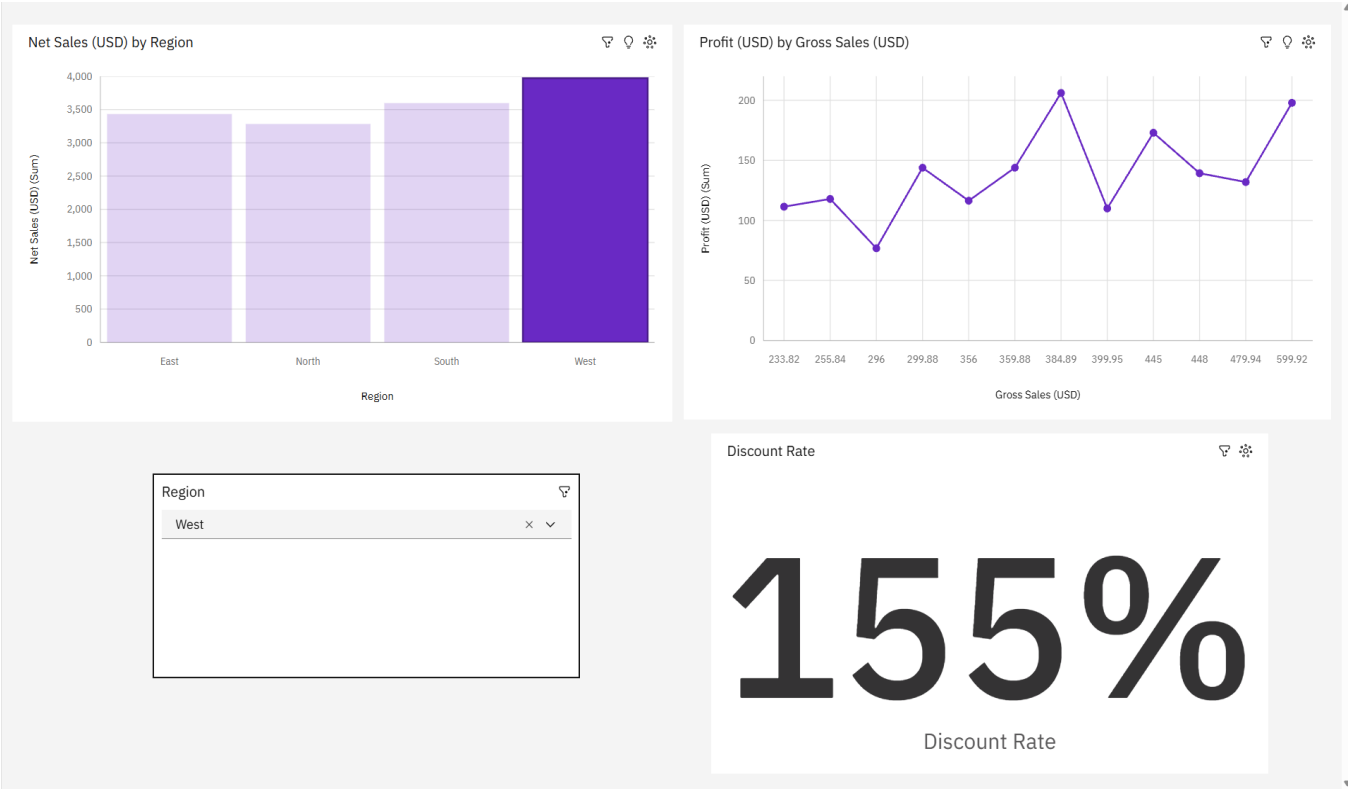
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6) Practical 6 : Build a Retail Pulse Sales Insights Dashboard

Objective: Assemble a dashboard combining a summary card, trend chart, segment breakdown, and a shared filter widget for Retail Pulse Sales data.







Result:- All widgets display live data with no placeholder values; the Region filter widget updates every other widget consistently;

the trend line renders clearly across the full period; layout follows a clear visual hierarchy with no overlaps.

Note on the Dataset:-

The Retail Pulse dataset is designed to analyze retail business performance and support data-driven decision-making. It contains 50 records covering sales transactions across different regions, stores, product categories, and supplier brands. The dataset includes key business indicators such as quantity sold, net sales, gross sales, profit, marketing expenditure, discount rates, and customer satisfaction. All monetary values are expressed in USD to maintain consistency in financial analysis.

This dataset is used in IBM Cognos Analytics to create interactive reports and visualizations, compare regional and product-wise performance, identify profitable sales patterns, and gain insights into customer behavior. It demonstrates how business data can be transformed into meaningful information to support better retail management and planning.

Link of Dataset Used for the Analysis:-

Retail_Pulse_Sales_Analysis_Dataset.xlsx
